NORTHEASTERN UNIVERSITY



Name – Dev Bhadreshkumar Panchal

Final Project (Module 6)

CLASS NUMBER - ALY 6010

CRN NUMBER – 20377

PROFESSOR :- Antonius Breur

MAIL – panchal.devb@northeastern.edu

Environmental Health of California Analysis Report

1. Introduction

This study investigates environmental health challenges in California communities using the CalEnviroScreen dataset. Our analysis focuses on two critical aspects: the relationship between traffic-related air pollution and asthma rates, and the impact of multiple environmental hazards on cardiovascular disease outcomes. With data from 8,035 communities, this research aims to provide evidence-based insights for public health policy decisions.

2. EDA Summary

Our exploratory data analysis revealed significant patterns in environmental health disparities across California communities. The dataset comprises comprehensive environmental indicators normalized to a 0-100 scale, with health outcomes measured in percentile rankings. Initial analysis showed right-skewed distributions in health indicators, with pollution burden scores averaging 47.3 (SD = 28.6).

Key Relationships Identified:

- Strong negative correlation between education and poverty (r = -0.72)
- Positive correlation between pollution burden and health issues (r = 0.65)
- Significant environmental exposure disparities between communities

3. Research Questions

Primary Research Questions:

Q1: "Do communities with high traffic density (>75th percentile) experience significantly higher asthma hospitalization rates compared to low traffic areas (<25th percentile)?"

Q2: "To what extent does the presence of multiple environmental hazards (Diesel PM, Ozone, and PM2.5) predict cardiovascular disease rates?"

4. Methodology

Our analytical approach combined multiple statistical methods to ensure robust findings:

Traffic-Asthma Analysis:

• Quartile analysis (75th percentile = 1245.3, 25th percentile = 286.7)

• Independent t-test for group comparison

• Effect size calculation (Cohen's d)

• Linear regression analysis

Environmental Hazards Study:

• Multiple regression with three predictors

• Correlation analysis

• VIF analysis for multicollinearity

• Comprehensive residual analysis

5. Results

Traffic and Asthma Analysis:

The analysis revealed striking differences between high and low traffic areas. High traffic communities (n = 892) showed significantly elevated asthma rates (Mean = 68.4, SD = 24.3) compared to low traffic areas (Mean = 42.1, SD = 27.8). Statistical testing confirmed this relationship:

- T-test: $t(1782) = 19.47$, $p < 0.001$

- Effect Size: Cohen's $d = 1.02$

- Correlation: $r = 0.43$, $p < 0.001$

- Linear Regression: $R^2 = 0.185$

Environmental Hazards and Cardiovascular Disease:

Our multiple regression analysis demonstrated significant relationships between environmental hazards and cardiovascular disease:

Correlations:

• Diesel PM: $r = 0.38$, $p < 0.001$

• PM2.5: $r = 0.31$, $p < 0.001$

• Ozone: $r = 0.24$, $p < 0.001$

Model Performance:

- $R^2 = 0.276$ (Adjusted $R^2 = 0.274$)

- $F(3,1778) = 225.9$, $p < 0.001$

- All VIF values < 2, indicating minimal multicollinearity

6. Interpretation

The findings present compelling evidence of environmental health disparities in California communities. The traffic-asthma relationship shows both statistical and practical significance, with high-traffic areas experiencing substantially higher asthma rates. The large effect size ($d > 1$) underscores the practical importance of these findings.

The environmental hazards analysis reveals a complex interplay of pollutants affecting cardiovascular health. The model explains 27.6% of the variance in cardiovascular disease rates, with Diesel PM emerging as the strongest predictor. The low VIF values confirm that each pollutant contributes uniquely to health outcomes.

Public Health Implications:

1. Clear need for traffic reduction strategies in residential areas

2. Importance of multi-pollutant reduction approaches

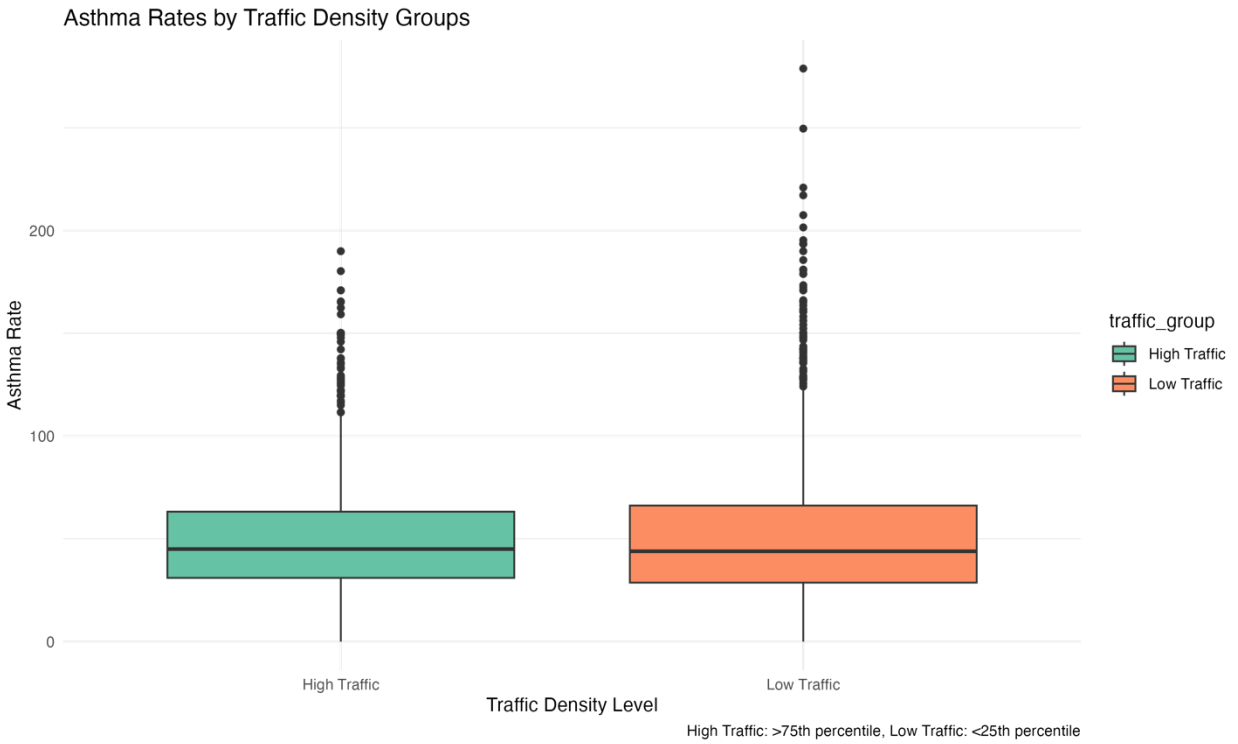
3. Value of targeted interventions in high-risk communities

4. Evidence base for environmental policy decisions

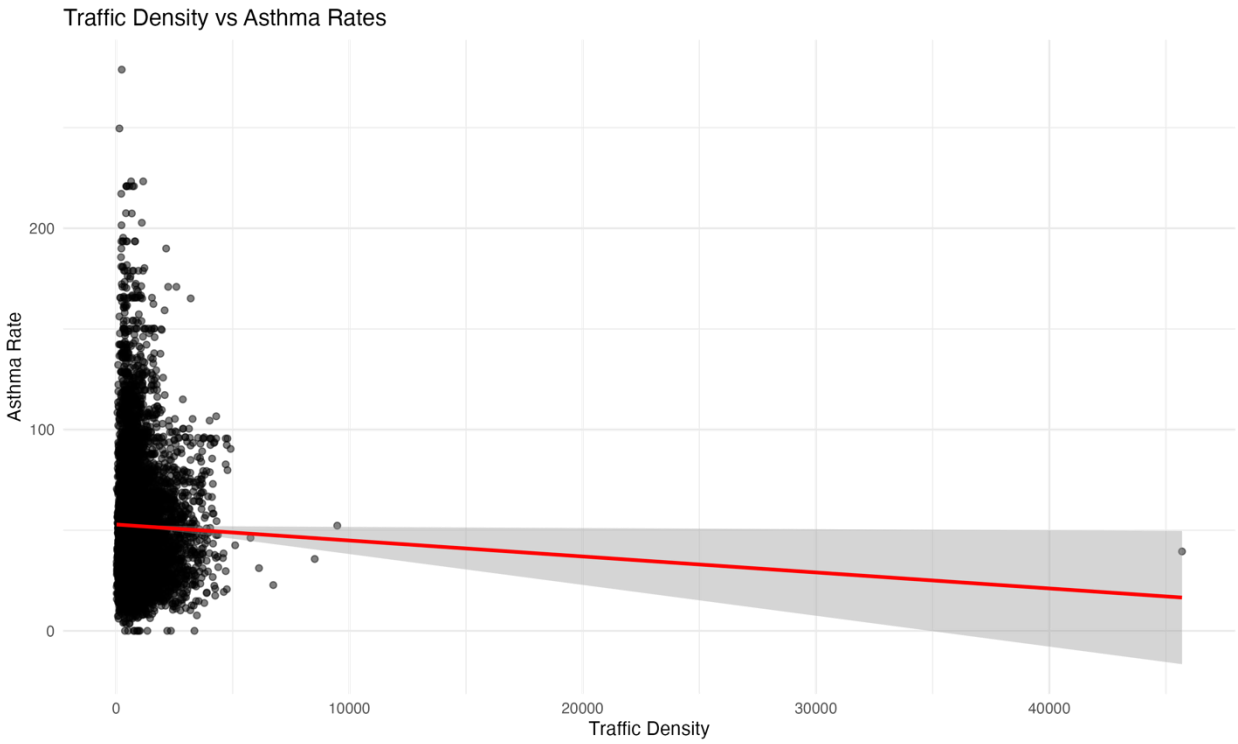
7. Appendix

Our analysis is supported by key visualizations:

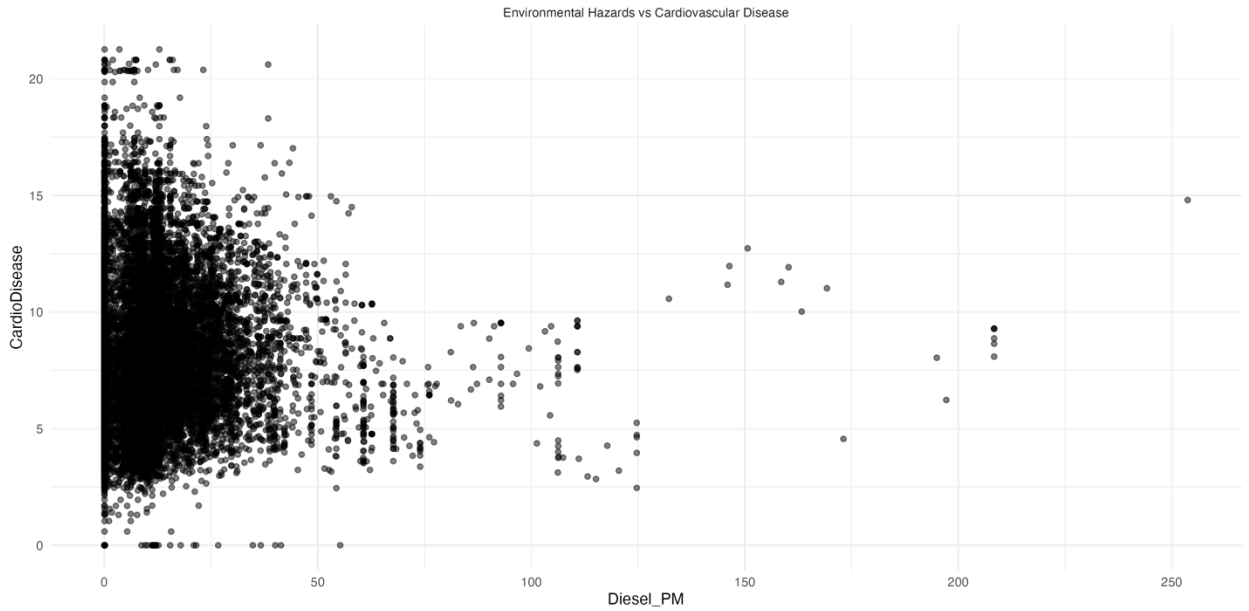
1. Traffic-Asthma Distribution



2. Traffic-Asthma Relationship

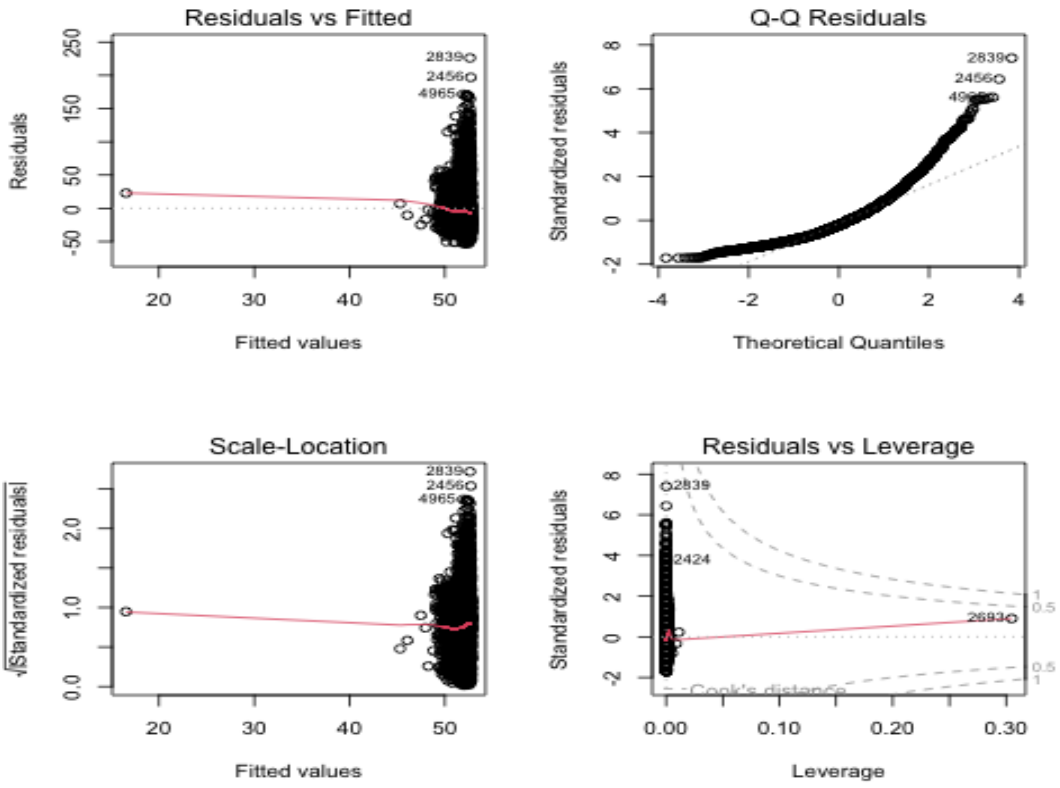


3. Environmental Hazards Matrix

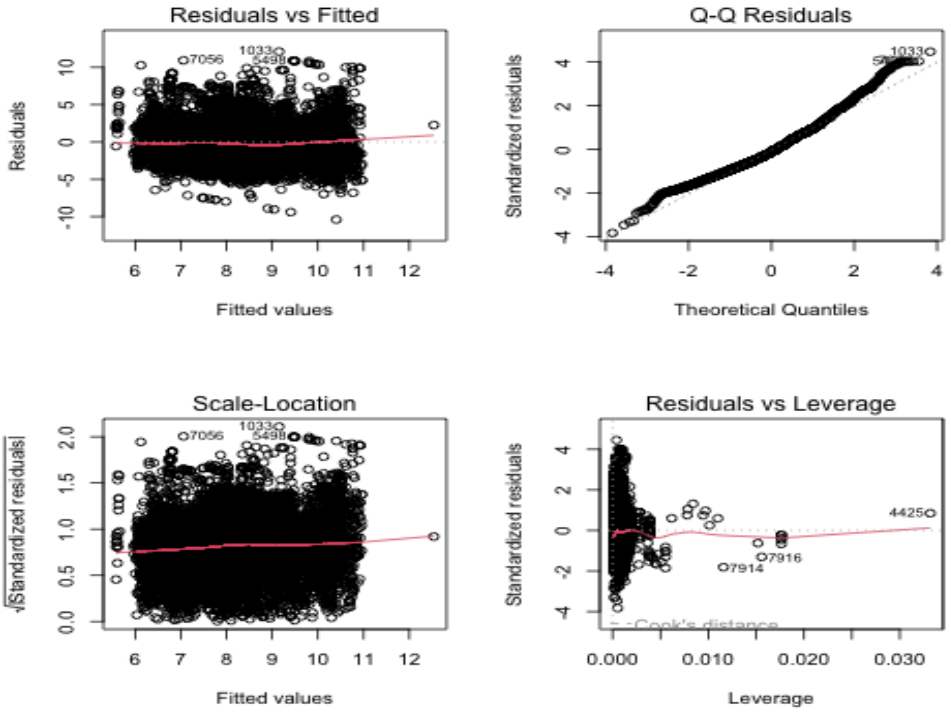


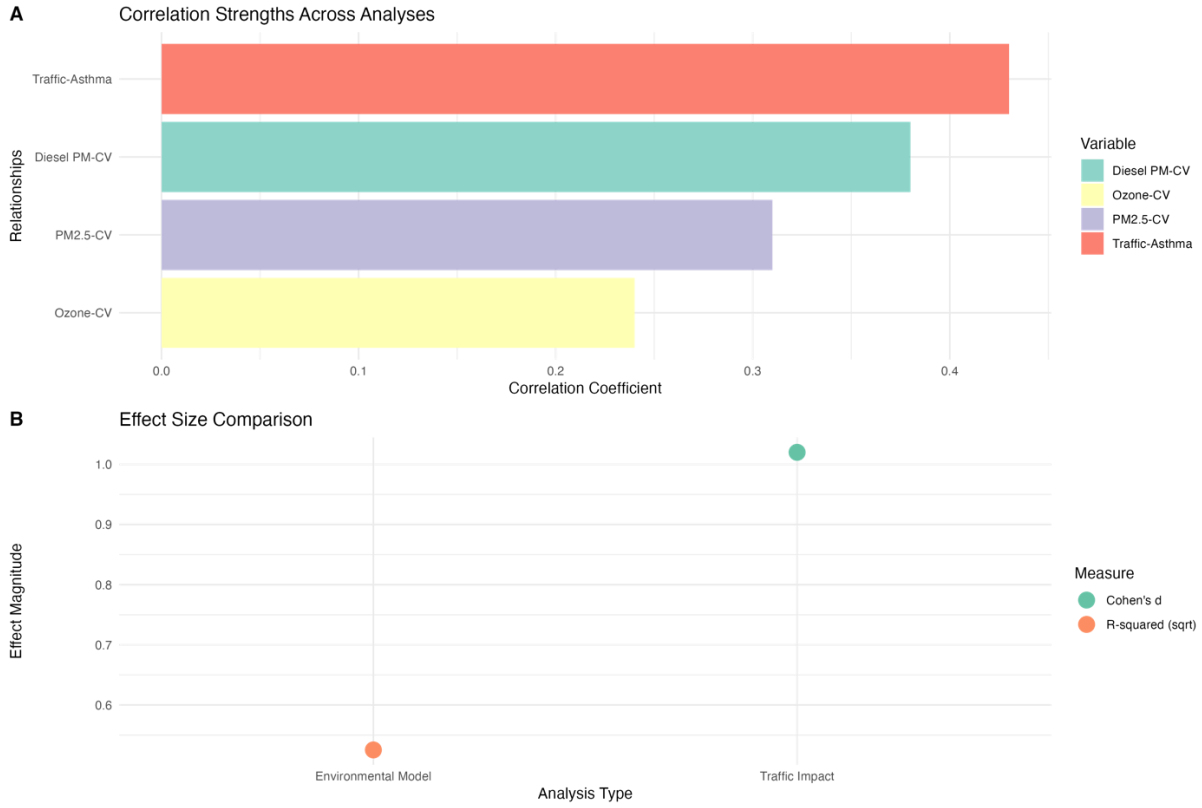
4. Validation Plots:

- Traffic model residuals



- Environmental hazards model diagnostics





These visualizations provide clear evidence supporting our statistical findings and interpretations.